

5G again

I'VE WRITTEN ABOUT MY OWN EXPERIENCE WITH 5G IN AUSTRALIA, WHEN I FIRST GOT IT A LITTLE OVER THREE YEARS AGO IN JULY 2019.

With India officially launching 5G services in September, I figured I'd write about my experience with the next generation of mobile services and see if any parallels can be drawn to what you should experience in India. Of course, Australia is the polar opposite of India. Australia has people spread out on a continent sized land mass with the highest concentrations being a few capital cities that have population densities of India's villages. While India is a large country as well, the population densities range from crowded to insane, and it's perfect for new mobile technologies.

First, let's fast forward through my three year experience of 5G. When it comes to my home broadband, apart from a configuration error of my closest tower meaning I was without any Internet for two weeks, because it happened during the pandemic and they couldn't get anyone out to fix it... I haven't really faced any noticeable down times to speak of. What that two weeks of having to use my mobile as a tether for my PC did was make me get a backup wired connection. And honestly, the wired connection goes down far more often than the 5G home broadband connection.

When it comes to 5G on my phone, while coverage has skyrocketed as compared to 2019, I still find a decent amount of teething problems and 5G dead spots – where my phone isn't handed over to a 4G tower despite 5G signal teetering on barely one bar of signal, and my internet experience feeling more like good old 3G! Despite my phone being on auto-select for networks, I can turn on Airplane mode and then turn the radios back on and immediately get full network strength 4G which feels orders of magnitude faster. I'm on a carrier that purchased the least bandwidth though, and which is perhaps using standalone towers (more on this later), so it could just be a problem faced by those using Vodafone in Perth, where I am.

Digit put out the first video of a 5G speed test on our socials, while we were at the Indian Mobile Congress – right after the PM announced that the 5G rollout had begun. One of our readers / followers made a comment that stuck out. He said something to the effect of, "Now we will finally get proper 4G speeds in the name of 5G". This is perhaps true, because while 4G is capable of speeds it has a problem with network congestion,

which is exactly what 5G aims to solve. Yes, it is true that I was getting similar speeds to the first Indian 5G tests on 4G here in Australia, four years ago, but Australia is a whole country that has less population than the much smaller area that is Greater Mumbai!

I expect India to speed ahead with the 5G rollout and for people to start getting average speeds of around 100-200 Mbps on their phones. This will mean much richer app experiences, the ability to game better, and an overall boost in connecting and downloading speeds across the board. Ping times will fall, so the ability to innovate with remote controls for various solutions will skyrocket. Once the bands of N78 (3.5 Ghz) and above start getting utilised properly, those are made for super high speeds and hugely congested networks. As the frequency rises the wavelength decreases, which means it becomes more prone to barrier interference. This is terrible for single towers covering large areas, but easily overcome with multiple small towers each covering a small area.

While there's no sense in Australian providers rushing to provide high frequency, high bandwidth 5G to the miniscule populations here, it makes total sense for the likes of Jio, Airtel and Vi to start setting up their higher frequency towers in India. Jio has already announced a plan to use standalone (SA) infrastructure. A bold move. Most carriers use non-standalone (NSA) towers, which basically means the towers are capable of sending out both 4G and 5G signals and switching people between them.

Eventually, however, the goal for all providers is to cover the whole of India with 5G SA towers and preparing for 6G to come in over the next decade or so. With 6G we'd expect an information superhighway that makes 5G's 20 Gbps peak speeds look like dial-up (remember that anyone?). With 6G everything will be connected – forget phones, we're talking cars, bikes, drones, robots, trains, buses, even electric cycles, traffic lights, streetlights... everything! I'm obviously getting carried away since 5G has just arrived!

Overall, very exciting times ahead for India! And networks are mere enablers. It's the way a billion innovative Indians will put the much higher bandwidth to use that really excites me. I cannot wait to see the next innovation cycle of our digital lives, and whatever those innovations are, they will surely be uniquely Indian! 🇮🇳



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